

Rakesh Kumar

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Education

Indian Institute of Technology (BHU), Varanasi

Bachelor of Technology in Computer Science and Engineering (CGPA: 7.93)

Dec 2021 – May 2025

Varanasi, Uttar Pradesh

Experience

Samsung R&D Institute India (SRI-Delhi)

Software Engineer – R&D (Core Platform Software)

July 2025 – Present

Delhi, India

- Resolved **120+ software defects (Kona)** and submitted **65+ changelists** via Perforce in a multi-threaded **C++** codebase. Investigated and fixed critical memory crashes (SIGABRT/SIGSEGV) and race conditions.
- Maintained the **Core Apps** platform and **AVControl** module, managing the deployment lifecycle from building **RPMs** to flashing and testing on target hardware.
- Collaborated across the Broadcast and Core Platform teams to manage cross-module communication using configuration APIs, ensuring smooth interaction between the hardware, UI, and accessibility layers.
- Developed an internal **Log Analysis tool** that automated the parsing and structuring of defect logs. Integrated an **MCP Server** exposing **REST APIs** for selective codebase retrieval and log filtering, connecting the pipeline to an AI assistant to accelerate root-cause analysis.

Exposure: C++, OOP, Multithreading, API Integration, Perforce, Debugging.

LTIMindtree

Summer Intern

May 2024 – July 2024

India

- Performed large-scale web scraping on Zauba Corp, extracting data of **2.7M+ companies** into MySQL.
- Developed a **Flask-based front-end** for streamlined database interaction and data visualization for the scraped data.
- Contributed to the frontend development of an **ML-based application** for extracting transaction details from payment screenshots.

Exposure: Python, Flask, BeautifulSoup, Render, GitHub, Cron-job, MySQL, HTML, CSS, JavaScript.

Skills

Programming Languages: C, C++, Python, Java, Dart, JavaScript, SQL.

Tools and Frameworks: Spring Boot, Django, Flask, Flutter, MySQL, Docker, Git, Perforce, REST APIs.

Core Concepts: OOP, Multithreading, Machine Learning, Collaborative Filtering, Web Scraping.

Projects

MyMoney | Personal Finance Tracker | GitHub | Website

Nov 2025 – Present

- Developed an offline-first personal finance application that allows users to monitor daily expenses, invite peers, and seamlessly manage and settle shared group ledgers.
- Engineered a **Flutter** app with **Firebase** real-time sync, **SQLite (Drift)** local caching, and **OAuth**. Optimized NoSQL schemas via subcollections and batched writes for scalable data reads.

Movie Recommendation System using KNN | Guided by Dr. Bidyut Kumar Patra | GitHub

Jan 2023 – Apr 2023

- Developed a user-based **KNN model** with collaborative filtering, leveraging **Cosine Similarity** to provide personalized movie recommendations based on historical ratings.
- Achieved final recommendations with **MAPE values of order 10^{-1}** using K-Nearest Neighbor model.

OTT Ratings | Guided by Dr. Amrita Chaturvedi | GitHub | Live

July 2022 – Nov 2022

- Developed an OTT Ratings platform using **Django**, allowing users to view, rate, and comment on web series including their seasons and episodes. Users can sort and filter series by preferences, or search directly by name.
- Utilized **Django models** to manage user details, ratings, comments, and web series data in the backend.

Honours and Achievements

Competitive Programming

- Maximum rating of **1469 (Specialist)** on Codeforces. Maximum rating of **1494** on CodeChef.
- Secured **Global Rank 50** in CodeChef August Long 2022 Div 4. Rank **1915** in Codeforces Round 852 (Div. 2). Rank **2199** in Educational Codeforces Round 142 (Div. 2).